Reg. No. | | | | | | | | | | |

[14EC62]

SIXTH SEMESTER B.Tech UNIVERSITY EXAMINATION, JUNE – 2018

INFORMATION THEORY AND CODING

Time: 3 Hours **Maximum Marks: 70**

Instructions:

Answer any FIVE full questions by choosing at least TWO questions from each part

PART-A

1. a. Consider the Markov sources whose state diagram is shown in the Figure. 1.a. Determine source entropy, G1,G2 and show that G1>G2. The state probabilities are P1=P2=1/2. **08**

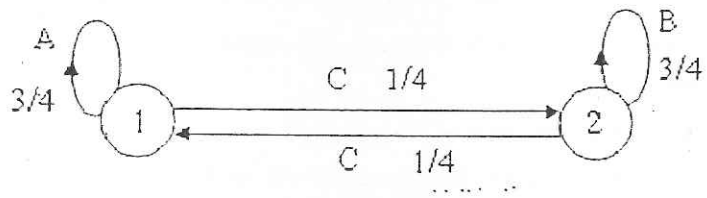

Fig. 1(a)

b. Consider discrete memory less source alphabet S={ S1,S2,S3} and source probabilities {0.5,0.25,0.25}.Calculate; **06**

- **i.** entropy of the source
- **ii.** Entropy of 2nd extension of the source.
- **iii.** From the above give your inference.

2. a. Consider a source with 8 alphabets S1 S2 S3, S4, S5, S6, S7, S8 with respective probabilities of 0.22, 0.20,0.18,0.15,0.10,0.08,0.05 and 0.02. **10**

- **i.** Construct a binary Huffman code and determine the code efficiency
- **ii.** Construct the ternary Huffman code and determine the code efficiency.

b. Determine whether the code given below is instantaneous or not? **04**

- **i.** By verifying Kraft inequality condition
- **ii.** By constructing code tree

Source symbols	Code
S1	1
S2	01
S3	001
S4	0001
S5	0100

3. a. For the channel whose matrix is given, p(x1) = 1/2, p(x2) =1/4, p(x3) = 1/4. Compute entropies H(x), H(y), H(x, y), H(x/y), H(y/x) and I(x,y) **08**

$$P(y/x)=\begin{bmatrix} 0.8 & 0.2 & 0 \\ 0.1 & 0.8 & 0.1 \\ 0 & 0.2 & 0.8 \end{bmatrix}$$

b. Prove with respect to mutual information of discrete communication channel **06**

i. $I(A, B) \geq 0$

ii. $I(A, B)=I(B,A)$

4. a. State and prove Shannon Hartley law on channel capacity. **07**

b. An analog signal has 4kHz bandwidth. The signal is sampled at 2.5 times the Nyquist rate and each sample quantized at 256 equally likely levels. All samples are statistically independent. **07**

i. What is the information rate of the signal?

ii. Can the o/p of the source be transmitted without errors over a Gaussian channel with bandwidth 50 KHz and S/N ratio is 23dB.

iii. What will be the bandwidth for transmitting the o/p of the signal without errors if the S/N ratio is 10dB.

PART - B

5. a. For a systematic (7,4) linear block code, the parity-check bits *c5, c6, c7* are formed from the following equations: **10**

$$C_5 = d_1 \oplus d_3 \oplus d_4; \quad C6=d_1 \oplus d_2 \oplus d_3; \quad C7 = d_2 \oplus d_3 \oplus d_4$$

i. Construct the generator matrix and parity check matrix

ii. Find the minimum weight of this code.

iii. Construct the encoder and syndrome circuit

iv. Suppose that the received word is [1001000]. Decode this received word by finding the location of the error and the transmitted data bits.

b. If C is a valid code vector then prove $CH^T=0$. **04**

6. a. For the (7,4) single error correcting cyclic code given $g(x)= 1+x=x^3$. Find the code vector for the message vector D=1101 using both nonsystematic and systematic method. **04**

b. Design an encoder for (7,4) binary cyclic code generated by $g(x)= 1+x=x^3$ and verify its operation using the message vector (1011). **05**

c. For the (7,4) single error correcting cyclic code given $g(x)= 1+x=x^3$. The received vector z(x)=1110101. Construct the syndrome circuit and determine the syndrome. **05**

7. a. Consider the binary convolution encoder shown in the Figure 7(a). Construct the state diagram, state table, state transition table and corresponding code tree. Using the code tree find the encoded sequence for the message (10111). Verify the output sequence so obtained using time domain approach. **14**

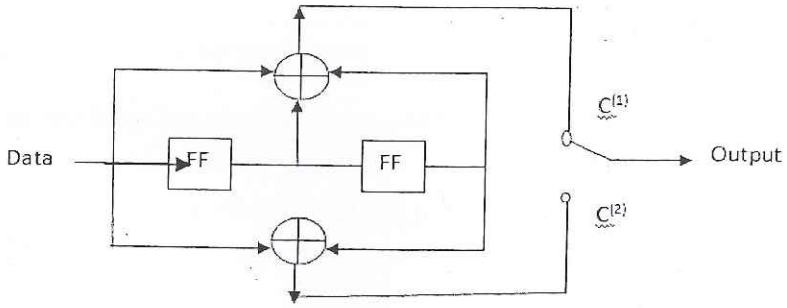

Fig.7 (a)

8. a. With a suitable example, explain any two of the following: **14**

i. BCH codes

ii. Reed Solomon codes

iii. Burst and random error correcting codes